NLP Research Project: From Research Paper to Independent Investigation

20% Continuous Assessment — 4th Year Undergraduate NLP

1. Core Philosophy

Students already know ML, DL, NLP fundamentals, embeddings, RNN/LSTM, Transformers, BERT, and basic experimental methodology. This project is therefore not a beginner assignment — it is a guided introduction to how research is actually conducted.

A research paper is an ANCHOR, not the project itself. Reproduction is the starting point; the student's own research investigation is the final contribution.

Intended research journey:

5-Paper Literature Survey → Select ONE Anchor Paper → Reproducibility Check → Baseline Reproduction → Identify Research Gap → Formulate Hypothesis → Predict Result Before Coding → Controlled Experiments → Ablation → Failure Analysis → Own Research Contribution → IEEE Paper + Defence

Students use the anchor paper to understand an existing research problem, reproduce a feasible baseline, identify gaps, consult related literature, and formulate their own research question.

2. Suggested Research Areas

Seven advanced, current NLP directions are offered as starting points; students are not restricted to these.

#	Research Area	Anchor Paper	Possible Student Directions
1	RAG Robustness & Reliability	Quantifying and Improving the Robustness of Retrieval-Augmented LMs Against Spurious Features in Grounding Data (ACL 2026)	Retrieval noise, irrelevant documents, ordering, reranking, grounding quality, answer robustness
2	Long-Context LM Training	How to Train Long-Context Language Models (Effectively) (ACL 2025)	Effect of context length, long-document understanding, truncation, relevant vs. irrelevant context, degradation
3	Long-Context Understanding	LooGLE: Can Long-Context LMs Understand Long Contexts? (ACL 2024)	Long-range dependencies, context length vs. performance, information position, retrieval vs. full-context reasoning
4	Query-Level RAG Robustness	Investigating the Robustness of RAG at the Query Level (2025)	Paraphrased queries, spelling noise, reformulation, synonym replacement, query complexity
5	LLM Hallucination	HALoGEN: Fantastic LLM Hallucinations and Where to Find Them (ACL 2025)	Hallucination vs. prompt/model size/temperature, domain-specific hallucination, RAG vs. non-RAG
6	Multilingual / Cross-Lingual LLMs	Cross-Lingual Pitfalls: Automatic Probing of Cross-Lingual Weakness in Multilingual LLMs (ACL 2025)	English vs. Indian languages, code-mixing, translate-then-reason, cross-lingual performance gaps
7	Synthetic Data for NLP	Understanding the Influence of Synthetic Data for Text Embedders (ACL Findings 2025)	Synthetic vs. human data, data volume/quality, generator LLM choice, low-resource NLP, embedding quality

3. Paper-Selection Rule

Students must NOT select one paper and start coding immediately. They must first identify at least 5 papers within ONE research area and compare them on: research problem, method, dataset, model, metrics, code/data availability, main finding, limitation, and possible extension. Only then is ONE Anchor Paper selected, with a written justification of why it is scientifically meaningful, (partially) reproducible, computationally feasible, and extensible into a research question.

4. Milestone Structure

Milestone	Component
M1	Literature Survey + 5 Papers + Anchor Paper Selection + Reproducibility Checklist
M2	Baseline Reproduction
M3	Research Gap + Original Hypothesis + Prediction + Experimental Plan
M4	Implementation + Controlled Experiments + Ablation
M5	Failure Analysis + Discussion + Limitations
M6	Final IEEE-Style Paper + Defence

M1 — Literature Survey

Submit: research area; five selected papers; comparison table; initial research problem; selected anchor paper with justification; and a reproducibility checklist covering:

- Public dataset available?
- Dataset size documented?
- Public code available?
- Public pretrained model available?
- Hardware requirement known?
- Training time known?
- Dependencies listed?
- Evaluation metric clearly defined?
- Feasible baseline runnable on Colab/Kaggle/free GPU?
- Are reported results sufficiently reproducible?

M2 — Baseline Reproduction

Reproduce a feasible baseline (not necessarily the entire paper) and document dataset, preprocessing, model, hyperparameters, training setup, metrics, original result, reproduced result, and the reproduction gap. Analyze the gap (seed, split, preprocessing, implementation, model version, compute limits) rather than hiding it.

A reproduction gap is evidence for analysis, not automatically a failure.

M3 — Research Gap + Original Hypothesis

Identify a gap from the anchor paper, related literature, and reproduction experience, then state ONE hypothesis with: (1) independent variable, (2) expected direction, (3) approximate expected effect, (4) mechanism/reason — written BEFORE running the experiment.

Format: "*Changing X will improve/decrease Y because Z. We predict approximately ___ improvement/change.*"

M4 — Experiment + Ablation

Core rule: change ONE major variable at a time. Report baseline, experimental configurations, predicted vs. actual result, quantitative metrics, comparisons, and ablations. Reward is for experimental reasoning, not merely a higher accuracy number.

M5 — Failure Analysis

Investigate where and why the approach fails — e.g., confusion matrices, per-class performance, hard examples, incorrect retrievals, hallucinations, long-context or language-specific failures. Provide ~5–10 representative failure cases, each addressing: What happened? Why might it have happened? What evidence supports this? Also report limitations, dataset/compute constraints, and threats to validity — with no unsupported high-stakes claims.

M6 — Final Research Paper + Defence

IEEE-style structure: Abstract, Introduction, Related Work, Anchor Paper and Reproduction, Research Gap, Proposed Hypothesis, Methodology, Experimental Setup, Results, Ablation Study, Failure Analysis, Discussion, Limitations, Conclusion, References. The paper must clearly separate what came from the anchor paper from what the student independently investigated.

5. Final Contribution Requirement

Reproducing the anchor paper alone is NOT sufficient for the final project.

Every project must include at least one meaningful student-led contribution: a new experimental condition, dataset/domain, robustness test, ablation, comparison, evaluation setting, error analysis, novel combination of existing ideas, investigation of a limitation, or challenge to an existing finding. For a 4th-year project, a well-designed empirical investigation is acceptable and desirable — it need not be a new model.

6. Marking Philosophy

Marks reward research thinking (not just coding), experimental discipline (not random trial-and-error), scientific reasoning (hypothesis → prediction → experiment → evidence → conclusion), critical analysis, honest reproducibility reporting, and originality beyond direct reproduction.

Higher accuracy alone does not mean better research. A negative result can receive excellent marks if the hypothesis, design, analysis and conclusions are strong.

7. Research Rules

1. Do not treat the anchor paper as the final project.
2. Do not copy the paper's experiments and call them your contribution.
3. Do not change multiple major variables simultaneously without justification.
4. Write predictions before experiments.
5. Report negative results honestly.
6. Use quantitative evidence.
7. Keep reproduced and original experiments clearly separated.
8. Do not claim state-of-the-art performance without valid evidence.
9. Use feasible models/datasets within available computational resources.
10. Cite all papers, datasets, models and external resources properly.
11. Do not make clinical, legal, financial or other high-stakes claims from classroom experiments.
12. The quality of reasoning matters more than the final accuracy number.

The goal is not to reproduce a paper. The goal is to learn how research is done by using a paper as the starting point for your own investigation.